

Title: Innovation Platform for Monitoring Health and Disease using Liquid Biopsies.

Acronym: (IPLiB)

1. General information

1.a Applicants

Name*	Institution	Department	Email	Tenured (i.e. fixed position/ Tenure-track/ Other	For Erasmus MC applicants:	Role in Flagship
John Martens	EMC	Medical oncology	j.martens@erasmusmc.nl	Prof. (100%)	Researcher	Main Lead
Marcel Reinders	TUD	Bioinformatics Lab	m.j.t.reinders@tudelft.nl	Prof. (100%)		Lead
Guido Jenster	EMC	Urology	g.jenster@erasmusmc.nl	Prof. (100%)	Researcher	Applicant
Pouyan Boukany	TUD	Chemical Engineering	P.Boukany@tudelft.nl	Assoc. Prof. (100%)		Applicant
Roland Kanaar	EMC	Molecular genetics	r.kanaar@erasmusmc.nl	Prof. (100%)	Researcher	Applicant
Sabina Caneva	TUD	Nanotechnology	s.l.caneva@tudelft.nl	Assist. Prof. (100%)		Applicant
Massimo Mastrangeli	TUD	Microelectronics	M.Mastrangeli@tudelft.nl	Assist. Prof. (100%)		Applicant
Menno Creyghton	EMC	Developm. Biology,	m.creyghton@erasmusmc.nl	Group leader(100%	Researcher	Applicant
Gijsje Koenderink	TUD	Bionanoscience	g.h.koenderink@tudelft.nl	Prof. (100%)		Applicant
Maarten IJzerman	EUR	Health Policy and Management (from 1-05-22)	maarten.ijzerman@unime-lb.edu.au	Prof (100%)		Applicant
Joost Gribnau	EMC	Developm. Biology	j.gribnau@erasmus.nl	Prof. (100%)	Researcher	Applicant

Corresponding Lead: John Martens

Planned start date of seed funding: 1-10-2022

Primary Convergence theme: *Technology Supported Transitions in Health(care)*

Secondary Convergence theme (if applicable): *Fundamentals of Health & Disease*

Tertiary Convergence theme (if applicable): Choose an item.

Quaternary Convergence theme (if applicable): Choose an item.

Keywords: liquid biopsy, microfluidics, dielectrophoresis, extracellular vesicles, single cell diagnostics, epigenetics, health care management

1.b Summary (292 of max 300 words)

Samples from blood, urine, saliva, ascites, bile, and cerebrospinal and nipple fluids are collectively referred to as “liquid biopsies”. They are considered to be extremely important diagnostic sources of biological information to assess health or disease conditions at any given moment in time with minimal burden for patients. Despite their potential, current technologies are lacking the sensitivity and specificity which are required to recover all critical information from body fluids. Therefore, technical advancement beyond the current state-of-the-art is urgently needed.

This Flagship converges experts in the fields of microfluidics, chip technology, single cell and single molecule phenotyping/sequencing, information sciences and modelling, and the clinical arenas (cardiology, microbiology, oncology, gastro-enterology, toxicology, internal medicine and neurology) to close the gap between fundamental research on ultra-sensitive detection methods and clinical application.

Our aim is to build a research community that uses their collective knowledge to develop novel advanced technology into standard operating procedures for minimally invasive health and disease monitoring.

Armed with these novel medical technologies, we will enable more (cost-)efficient health and disease monitoring in both individual patients as well as for risk management of the general population. This will allow more effective clinical disease management even before obvious signs of clinical disease are manifest. Furthermore, our approaches will provide more detailed and specific insight into the onset and progression of diseases. This will allow us to discover a wide range of clinically relevant health and disease states that can be leveraged to identify risk populations and provide better and more personalized intervention strategies for all patients. The discovery and clinical potential of such novel biomarkers will also provide an unique opportunity to attract new sources of funding, including private-public partnerships, to come to a sustainable Flagship on fundamental and translational liquid Biopsy research.

1.c Public Summary (247 of max 300 words)

Vloeibare biopten (bloed, urine, speeksel, ascites, hersenvocht, enz.) zijn in potentie belangrijke diagnostische informatiebronnen, omdat ze op elk moment met een minimale belasting voor het individu of de patiënt afgenomen kunnen worden om de gezondheidstoestand van een individu te kunnen beoordelen. Echter, het verkrijgen van nuttige gezondheids-gerelateerde informatie uit lichaamsvloeistoffen vereist ultra gevoelige en uiterst specifieke testen. Aangezien dergelijke detectiemethoden op dit moment veelal niet beschikbaar zijn, is er dus technologische vooruitgang nodig voordat we de potentie van vloeibare biopten klinisch ten volle kunnen benutten.

Om de technologische kloof tussen belofte en toepassing te dichten, werken in dit *Flagship* experts op het gebied van microfluidica, chip technologie, moleculaire fenotypering en informatiewetenschappen samen met deskundigen uit relevante klinische vakgebieden als cardiologie, toxicologie, oncologie, microbiologie, interne geneeskunde en neurologie, om middels fundamenteel onderzoek nieuwe ultragevoelige detectiemethoden te ontwikkelen en hun klinische toepasbaarheid te toetsen. Ons doel is om een netwerk van onderzoekers op te bouwen dat gezamenlijk geavanceerde technologieën ontwikkelt op het gebied van minimaal invasief monitoren van gezondheidstoestanden.

Ons einddoel daarbij is om kostenefficiënte monitoring van gezondheid en ziekte mogelijk te maken bij zowel patiënten als de algehele bevolking. Dit maakt interventies mogelijk voordat fatale ziekten zich manifesteren of voordat ziekteprogressie is opgetreden. Bovendien verwachten we dat de ontwikkelde gevoelige meetmethoden beter inzicht verschaffen in ontstaan en progressie van ziekte en helpen bij het ontdekken van nieuwe gezondheid gerelateerde biomarkers die gebruikt kunnen worden om risicopopulaties te identificeren en betere en meer gepersonaliseerde interventiestrategieën te bieden of te ontwikkelen.

2.Relevance to Convergence Health & Technology mission

(50%) — max. 2500 words for 2.a to 2.e

2.a Contribution to H&T mission and fit with H&T theme(s)

Liquid biopsies have a wide variety of applications relevant in early detection of a host of diseases, even before symptoms become overt, thus allowing preventive or curative measures to be initiated early-on. Therefore, almost by definition, this innovative field of diagnostics contributes to a value-based & affordable healthcare system. Our Flagship's aim is to radically improve state-of-the-art liquid biopsy analysis technologies and therefore entirely fits the main Convergence H&T mission, which is to promote and maintain health for all. Our aim is to develop revolutionary new molecular instrumentation technologies and data analysis methods for the analysis of liquid biopsies in a variety of health applications, which most significantly contributes to the Convergence theme "Technology-based transitions in health(care)". Since liquid biopsies provide an untapped window to determine the molecular status of disease and its progression, our Flagship will also contribute to the "Fundamentals of Health & Disease" theme of Convergence, which aims to understand the molecular basis of life and disease processes across the scales ranging from molecule to patient.

2.b Synergy between disciplines

This Flagship connects translational (Martens/Jenster) and fundamental researchers (Gribnau/Kanaar/Creyghton) from Erasmus MC with various technical disciplines from TU Delft (Pouyan/Mastrangeli/Koenderink/Caneva/Reinders) to deliver sophisticated and innovative devices with integrated detection and analysis tools to aid, together with the EUR, value-based clinical decisions in health care (IJzerman) by using well-defined diagnostic markers in body fluids. The combination of isolation of disease-derived cells (Martens/Boukany/Mastrangeli) and extracellular vesicles (Jenster/Caneva) and a novel technology to fingerprint nucleic acids (Martens) and associated epigenetic modifications (Creyghton), and epitranscriptomic (E. Martens) as well as proteins and protein modifications at the single-molecule level (Caneva) is expected to transform the field of non-invasive biomarker development in a value-based manner (IJzerman). To realise the potential of analysing bodily fluids for health care purposes, it is mandatory to build a transdisciplinary community of scientists in the fields of cell biology (Gribnau/Kanaar/Creyghton), biophysics (Boukany/Koenderink), microfluidics (Mastrangeli) and ultra-sensitive phenotyping (Caneva) as well as data analysis and integration (Reinders) devoted to gathering novel knowledge from body fluids. While various parties within the Flagship are collaborating in a bilateral fashion, uniting them in this Flagship will deliver a flywheel to advance the field. To close the gap between ultra-sensitive detection methods developed in this Flagship and clinical application, affiliated parties from cardiology, microbiology, oncology, toxicology, internal medicine and neurology (listed Ad 4c) are on board to implement our methods, as these have wide applicability to various disease

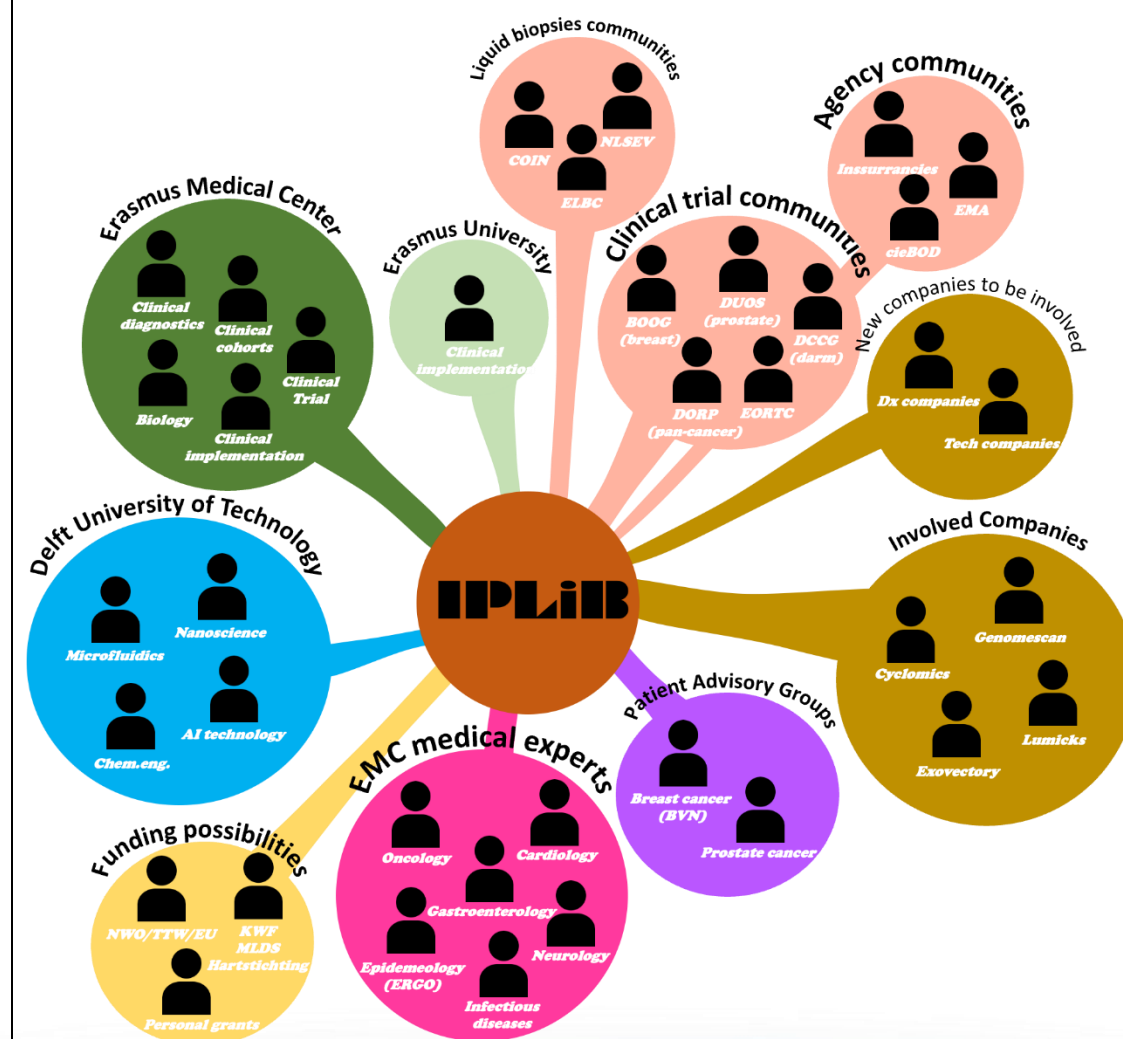
states including early detection, prognostication, response prediction and disease monitoring as well as studying disease evolution over time. Our multidisciplinary approach is essential to bring advancements emerging from our technologies from bench to bedside in a cost-effective manner.

2.c Long term sustainability plan

Sustainability of our Flagship is dependent on three pillars:

1. Community building (collaboration & networking),
2. Resources (acquisition of funding; valorisation of output of the project)
3. Education (Scouting talent and educate next generation transdisciplinary scientists)

Our **first pillar** is building a community where all parties involved intensively collaborate and effectively interact with relevant external parties. To work together within the Flagship, we will organise joint monthly meetings per WP as well as quarterly engagements for the entire Flagship. A digital workspace (IPLiB) has already been created to stimulate joint efforts. To stimulate interaction with external parties, we will organise annual scientific hackathons with suitable affiliated parties invited along with the core consortium members to discuss future directions and novel opportunities. Additionally, as also mentioned in the management paragraph (4.d), the annual Flagship progress meeting will be held with SAB present as well as external users of the developed technology, which will help to strengthen and expand our community (diagram below).



As soon as our technology is mature and can be moved towards clinical application in a value-based manner, we will involve affiliated clinical collaborators (listed in 4.c) for feasibility studies. At this moment patient advocates (PAGs) will also be consulted. We have ample experience in involving PAGs (e.g. those from “Borstkanker vereniging Nederland”) in evaluating innovative clinical/translational research. For actual implementation, we are connected to the COIN initiative (headed by prof Gerit Meijer, Netherlands Cancer Institute), which aims to implement liquid biopsy assays into clinical disease management. Thus, our Flagship has everything lined up to bring our developed technology into the clinic, which further adds to our sustainability. Lastly, on an international level, we are connected to European Liquid Biopsies Society (ELBS) to implement research outcomes on a European level.

Our **second sustainability pillar** is to generate resources. On the one hand this involves, based on the output of our activities, attracting funding for our research from charities (KWF/MLDS/hartstichting, etc), the national organisation for scientific research (NWO) and European funding bodies. Therefore, this Flagship will fuel bilateral grant applications between Flagship partners and our affiliates as well as seek funding as a community within national or international programs (TTW Perspectief/NWO Gravity/NWA/European Horizon program). Also personal grant applications (Vidi/Vici/ERC) will be supported since our unique transdisciplinary Flagship will benefit its individual members.

Apart from applying for funding, we will make sure to have liquid biopsy research and related technological development profoundly present on National and European Scientific agenda's and on those of charity foundations. Importantly, the Dutch cancer society has already made disease prevention, and thus implicitly liquid biopsy research, a priority, but we will make sure it will also become a vested interest in other disease-related funding agencies.

Another important factor for generating resources is valorisation, including IP protection. Our aim is to bring devices and diagnostic assays to demonstration level (TRL6) after which commercial parties will be engaged or start-ups initiated, which will be another sustainable opportunity to create resources. Additionally, we intend to set up partnerships with companies. LOIs have already been provided by Cyclomics, Lumick and ExoVectory and we expect others to join. Furthermore, parties in the consortium actively collaborate with important players in the field of Liquid biopsy research (listed in 4c). Another sign of the Flagship being able to sustain itself with regard to generating resources is past performance. This consortium is currently involved in translational research and technology development in Flagship related research for ~0.75-1M euro annually, so we expect parties involved to continue to secure grants from public and private funding for the entire project period. In fact, we anticipate that by forging synergies across disciplines, this Flagship will sustain itself without further central support to allow convergence on this topic for the next decade.

The **third pillar** of a sustainable future is to educate the next generation and engage young talented scientists to carry forward this emerging transdisciplinary field at the interface of technology and biomedicine. Within the Flagship, we have already pre-sorted the next generation of mathematical (dr. Stavros Makrodimitris, TUD), translational (dr. Saskia Wilting, EMC) and clinical researchers (dr. Nick Beije, EMC). Thus, this Flagship has already made important steps to be sustainable beyond the timeframe of this Flagship proposal. More details on how we aim to attract talents are outlined in Ad. 2d.

Finally, as past performance is a good measure of success, we have previously delivered liquid biopsy-based tests or biomarkers in general (listed below), further supporting our capabilities to deliver clinically relevant and implementable results.

For instance:

- Outcome prediction based on PITX2 DNA methylation (IVD test; Therawis)
- Prognosis and pathway predictors from gene expression data (IVD test; Philips and J&J)
- Predictive value of variant androgen receptor (ARv7) on CTCs/exosomes.
- Urinary EV assays (e.g. EVQuant).

To conclude, since founding partners of this Flagship have proven themselves able to bring this field forward individually, their convergence almost guarantees important progress will be made.

2.d Visibility, attracting talents and new partners

To become visible within the founding universities of convergence, we will use internal media such as “Amazing Erasmus” and “Rising Stars@TUD” to present ourselves. On a national level, we will exploit opportunities created by NWO such as “Bessensap” to present research to society. Also we will make an effort to present ourselves on national TV in scientifically oriented programs such as ATLAS. On an international level, we will broadcast our activities via LinkedIn, twitter or instagram. Within the scientific community, we will continue to collaborate with national scientific committees such as the Netherlands Society for Extracellular Vesicles (NLSEV) and the COIN consortium. Furthermore, the PI of the consortium is currently creating synergy nationally through an NWA application, vais the mentioned ELBS and clinically within the EORTC. As a result of these activities, we expect our Flagship to create visibility for the general public as well as broadly within the scientific community.

To attract talent, we have several already ongoing activities. The first is the Junior Med School program in which Erasmus MC aims to attract the best doctors-to-be already at high school level. We have contributed since the start of this program (2006) a topic on liquid biopsies and have delivered last year the first PhD on the liquid biopsy research, dr Lindsay Angus, attracted via this program and shows we are already attracting talent early-on. Additionally, we aim to retain talent. Therefore, we will set up a “scout and behold” program within the Flagship. This program actively engages talented undergraduate students from TU Delft, EUR and Erasmus to the Flagship and tries to keep them engaged while they progress during their studies, creating a pool of talented students whom we will offer MSc and BSc projects and stimulate to continue in a PhD project to be created for/with them.

With regard to education, apart from the junior med school program already mentioned, liquid biopsy research is currently part of an elective course within the Nanobiology curriculum (TU Delft/Erasmus MC) and a minor for medical students (Erasmus MC). The new graduate school biomedical sciences offered to deliver a program on Liquid biopsy research for our PhD students (EUR/EMC). Liquid biopsies research is also part of the training of health care professionals of the Netherlands organisation for oncology (NVvO). It is our intention to make educational programs at all stages of the academic development of our students. Finally, exchange programs to be set-up with other European universities (e.g. Graz, Hamburg) will allow PhD students working within the Flagship to further advance themselves and allow them to deliver knowledge to our Flagship in return. Our aim is to deliver a steady inflow of students from all the necessary disciplines to let the Flagship flourish even further.

With regard to attracting partners, the most important partner to attract first is the EUR. EUR is modestly represented,, probably due to the fundamental and rather technological nature of our Flagship. We expect that as soon as the first tools are delivered clinically or commercially,

involvement of EUR will be straightforward, especially in the domain of ethics and health economics and policy.

The Flagship in its current setup is oriented mostly on cancer research, as this is the first field where liquid biopsies have shown their clinical potential and is a one case already part of routine diagnostics. Our aim is to go beyond the field of cancer and many parties have shown interest in this Flagship to improve diagnostics and treatment of patients in additional disease categories (see list 4b. In fact there are already ongoing collaborations of members of the Flagship with scientists in the field of toxicity, forensics and neurodegenerative disease. Indeed since our Flagship focusses on developing innovative technology, we expect broad application within many different areas of health and disease. A third community we need to drill into is companies working in the field of liquid biopsy research. We have contacts with several parties in the field (listed in 4.c) who we will include more actively representing an activity of the Flagship under WP5 (public-private partnership). The fourth community is the national and international scientific community, where we aim for all members of the Flagship to be engaged and to actively interact, e.g. by building an ITN network with members of the ELBC and beyond.

In sum, we have developed a comprehensive community of participants and partners from both technical and medical research and health care development with a healthy influx of new talent and with substantial national and international visibility.

2.e Long term societal impact

The aims of our Flagship are to develop innovative ultra-sensitive and specific molecular tests and data analysis tools and devices, to analyse liquid biopsies of any kind with the mission to improve health care in a patient friendly and value-based manner. We expect the products to be developed to increase our fundamental understanding of how rare analytes in body fluids harbour information about the individual's health status and to provide value as diagnostic tests to maintain health or monitor disease.

Better disease staging and (therapy) monitoring through increased clinical utility of liquid biopsies will improve disease management of patients. However, these technologies will also enable more efficient health monitoring in the general population, allowing prevention and early detection diagnostics to proactively maintain health. Specifically, we aim to make liquid biopsies more efficient, allowing better detection of current biomarkers but also develop new biomarkers that detect disease or pre-disease stages. Both of these aspects of better health monitoring (for patients and healthy individuals) benefit lifelong health from pre-birth to end of life. Scale up of these systems to the general population or to those suffering from a specific disease will provide insight into the heterogeneity of these biomarkers within the relevant populations. This will benefit the future development of personalised treatment strategies fitting individual-tailored precision medicine for all. Liquid biopsies carry the promise of affordable, patient-friendly health monitoring, and thus are promising alternatives to contribute to cost-effective ways to detect and diagnose diseases. We will include value-based healthcare approaches in the assays we develop from early on. This will make our healthcare system more affordable and, by removing thresholds for screening and diagnosis, contributes to socio economic equality in health care.

Apart from the direct effect on individual health, the project also delivers products with economic value.

We specifically expect 3 types of economic output:

1. The proof-of-concept devices to enrich rare analytes (cells, extracellular vesicles, particles, etc) to be developed are licensed to existing or diagnostic or startup companies (WP1/2).
2. Ultra-sensitive and specific assays to analyse rare analytes to be developed are licensed to diagnostic (startup) companies (WP3).
3. Computational algorithms to analyse high-throughput data from the various assays (WP4) will be made available as a service or as a product to be out-licensed to companies or integrated in a device being developed.

As a result of these deliverables the Flagship will stimulate economic activity of biotech to bring these to the market, creating jobs and thus prosperity, which will benefit society in the long run.

3.Flagship proposal (25%) max 2000 words for 3.a to 3.d

3.a Ambitions

In the era of precision medicine, liquid biopsies are considered the holy grail for monitoring health and disease over time. The promise of liquid biopsies is that, by obtaining a minimally-invasive sample from a body fluid, representative health-related characteristics can be determined in real time and repeatedly, thus enabling personalised health and disease monitoring at any given moment [1,2]. This body fluid is often blood, but also other body fluids such as urine, saliva, seminal fluid, bile, sweat, liquor, ascites, etc , provide representative information about a tissue or organ of interest while requiring minimal or moderately (for bile, liquor, ascites) invasive techniques. The current landscape of types of liquid biopsies comprises counting and characterization of rare cells, e.g. circulating tumor or endothelial cells, or analysis of nucleic acids directly derived from plasma (so called cell-free DNA (cfDNA)). Emerging technologies in liquid biopsy analyses include analyses of cell-free RNA, extracellular vesicles (EVs), such as exosomes and microvesicles, circulating proteins and metabolites, and educated platelets [1,2].

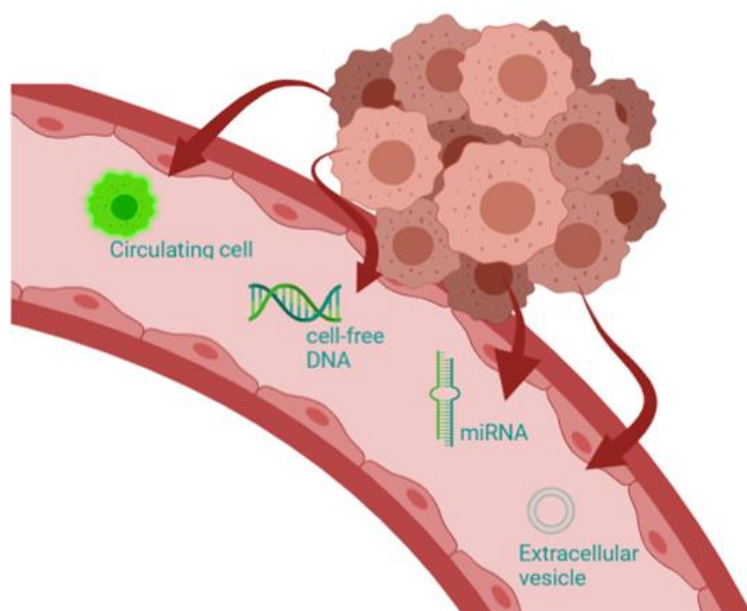


Figure 1. Examples of health and disease related biomarkers in blood, a commonly used liquid biopsy.

However, the clinical utility of liquid biopsies is still rather scant, with a few notable exceptions, such as monitoring cancer via cancer antigens including prostate specific antigen (PSA) [3] and detection of resistance mutations in lung cancer during EGFR-targeted therapy [4] and outside the field of cancer in non-invasive prenatal testing [5]. The reason for the limited number of applications is that current liquid biopsy-derived assays still have important caveats. The biggest challenge relates to the fact that biomarkers to be analysed in liquid biopsies are often extremely rare and detection thus requires the use of very specific methods and/or to selectively enrich analytes of interest which demands, even after enrichment, for very sensitive methodology to raise the signal of interest above background noise. This Flagship unites experts in the field of microfluidics, micromanipulation/sorting, chip technology, single cell-omics and -epigenomics, single EV and (macro)molecule phenotyping/sequencing and data analytics and data integration to break down the wall that currently prevents clinical utility of liquid biopsies for improvements of health care cost-effectively.

3.b Aims and feasibility

(270 words)

Hypothesis. Liquid biopsies are pivotal to understand and monitor disease and retain good health.

Objective: Our objective is to build a scientific community that develops robust technologies to collect accurate and critical predictive information from rare molecules, organelles, and cells of interest present in body fluids.

Organisation: The Flagship is organised according to the following work packages:

WP 1. To develop methodologies to purify and phenotype rare diseased cells from body fluids.

WP 2. To improve methodologies to phenotype individual extracellular vesicles from body fluids.

WP 3. To develop multi-omic assays.

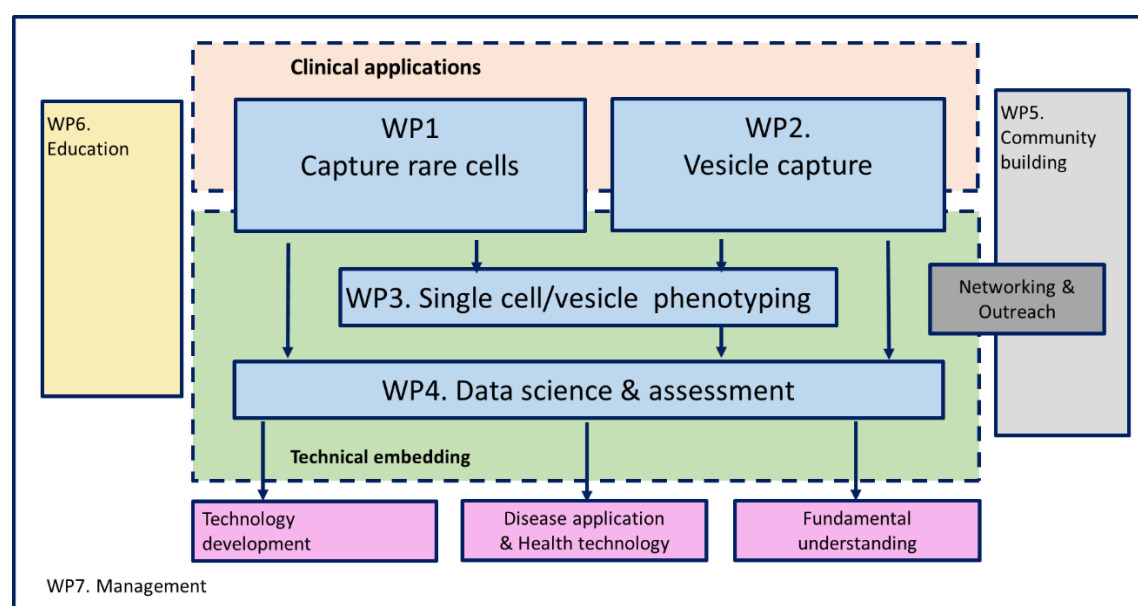
WP 4. To develop pipelines to interpret data generated by liquid biopsy assays.

WP 5. To build a private-partnership community.

WP 6. To develop educational programs for the next generation.

WP 7. To manage the Flag, to valorize results and reach out to society.

The Flagship diagram and its objectives. Each WP will focus on cancer as well as another disease.



Expected outcome. We consider our project successful if we can deliver two prototype devices (TRL6 after 3 to 5 years) and methodologies to enrich and purify cells, extracellular vesicles, and biomolecules (nucleic acids/proteins) from body fluids, as well as assays and software to characterise them. One prototype involves the development of microfluidic devices for the isolation and functional and molecular characterization of rare cells; the other a novel acousto-photonic nanopore platform for single protein fingerprinting of marker proteins from disease-derived EVs. We will assess the socio-economic value of the solutions we develop from early on, to make sure they are also cost effective as well and thereby beneficial to society.

3.c Work plan

WP1. To develop methodologies to purify and phenotype rare diseased cells from body fluids.

Participants: Boukany (lead), Martens, Mastrangeli, Koenderink, Kanaar.

Task. We will develop a microfluidic device for biomarker-independent cell isolation and subsequent functional and epigenetic single cell phenotyping.

Requested: one PhD student; a part-time technician

Rationale: Our main aim of this multidisciplinary task is to develop a label-free microfluidic cell sorter [6] to capture and (functionally) characterise rare diseased cells from peripheral blood of patients diagnosed with a specific disease. Initially, we focus on solid tumours but aim to expand to other diseases (listed in 4c).

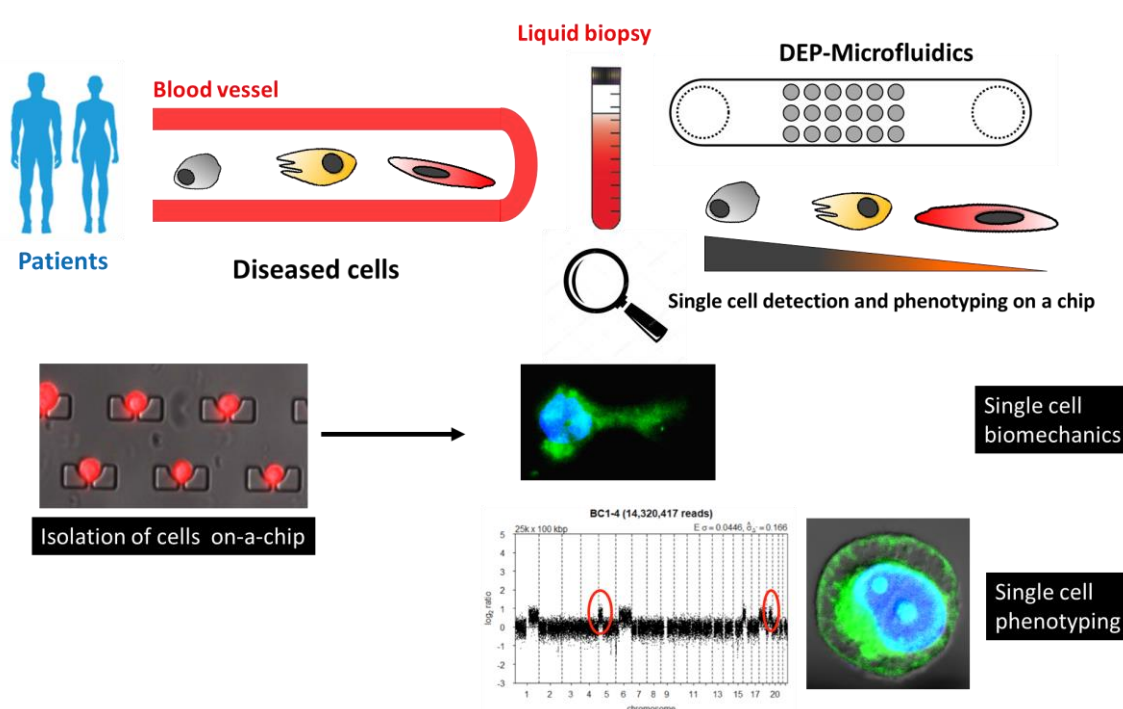


Figure 2. Overview of WP1 involving single cell recovery from liquid biopsies and subsequent biomechanical (WP1) and molecular phenotyping (WP3) .

Task 1.1: We will employ a novel label-free microfluidic technology to selectively capture without cellular loss during processing rare cells (ranging from 1 to 1000 per tube of blood; spike in initially; later cases patient blood) by taking advantage of the biophysical and biological properties of the aberrant cells. This dielectrophoresis (DEP)-based cell sorter will be integrated with advanced micro-electronics to isolate and manoeuvre single diseased cells in a controlled manner for further characterizations of single cells on-a-chip.

Task 1.2: By combining DEP-based cell sorting with a single-cell microfluidic assay for efficient characterization of rare cells on-a-chip, we will be able to phenotype key biophysical features of captured single cells at the cellular and sub-cellular level (e.g. single cell nuclear deformability; DNA

repair deficiency/RAD51 foci formation [7], apoptose, senescence [8], etc). Furthermore, captured events will be delivered to WP3 for analysis of cellular identity using single-cell single cell (epi)genomics.

WP2. To improve methodologies to phenotype individual extracellular vesicles (EVs)/lipoproteins from body fluids (focus initially on cancer and atherosclerosis/dyslipidemia).

Participants: Jenster (Lead), Caneva, Boukany, Mastrangeli, Koenderink.

Task. We will develop an acousto-phonic nanopore platform to phenotype individual EVs.

Requested: one PhD student, a part-time technician.

Rationale. To communicate, all cells secrete EVs from the host cell containing disease-related content if derived from a diseased tissue. Isolating physical characteristics of individual EV (instead of bulk) is an unexplored field of research [9] and is hampered by technological challenges related to their small size (50-200 nm), low abundance (as compared to 'normal' EVs) and low levels of disease related biomarkers.

Task 2.1. Single EV detection technologies that will be further developed include:

- microfluidics-based capture EVs (using specific antibodies) or, similar to WP1, using unique biophysical properties,
- Confocal microscopy of fluorescently labelled ZVs using the EVQuant platform [10]
- Mass photometry by Interferometric Scattering Microscopy (iSCAT) [11].

Task 2.2. Technologies employed for the molecular characterization of the disease-associated EVs include:

- single-molecule protein fingerprinting with solid-state nanopores/nanochannels[12],
- disease-associated RNA detection at single EV level using digital RT-PCR [13],
- selecting nanobodies specifically binding disease-derived EVs using next generation phage display technology.

To show proof-of-concept, EV detection and molecular characterization assays developed will be tested on urine and plasma samples for the diseases of interest.

Task 2.3: When successful will apply the novel technology in cardiovascular disease to selectively capture rare lipoproteins by taking advantage of the biophysical and biological properties and protein exterior of these lipoproteins.

In direct collaboration with WP4, the biophysical and molecular data collected in this WP will be integrated via computational methods as outlined in WP4.

WP3: To develop mono and multi-omic assays

Participants: Creighton, Gribnau, Boukany, Kanaar, Koenderink, Caneva.

Requested: one PhD student

Rationale: Cell identity as well as cell state is determined by epigenetic control of promoters and enhancers that determine the gene expression programs, which in turn dictate cell identity [14]. Measuring disease state from body fluids thus requires robust identification of disease-associated alterations to the epigenome and (epi)transcriptome in small quantities of input material.

Task. We will adapt the epigenetic and epitranscriptomic profiling technology to obtain multi-omic data from cells isolated based on their biophysical properties in WP1, from vesicles isolated in WP2 and from bodily fluid directly (this WP). Data analysis will be done in WP4. A reference library of disease-associated cell types with epigenomic and (epi)transcriptomic data directly relating biophysical properties from WP1/2 to multi-omic data will be realised, which may enable the prediction of disease state based on biophysical properties only [15]. Biophysical phenotyping would greatly enhance the throughput and deployability of such assays in the clinic.

WP4. To develop pipelines to interpret data generated by liquid biopsy assays.

Task. We will build bio-informatics/computational algorithms to integrate and interpret data from innovative platforms and assays developed in WP1, WP2 and WP3.

Participants: Reinders (Lead), Gribnau, Martens, Jenster, IJzerman.

Requested: one Post-doc and part-time senior scientist

Rationale. As detection of targets within liquid biopsies suffers from the underrepresentation of these targets, we will opt for two different strategies to optimise readout, 1) detecting suitable biomarkers, and 2) decreasing noise levels for selected biomarkers. Moreover, having omics-based biomarkers, we aim to translate these into physiological markers, to achieve simple and high throughput screening of targets using our microfluidic device(s).

Task 4.1. To steer the selection of omics biomarkers, we will follow a hybrid machine learning approach, in which public data and prior knowledge are used to steer the architecture of a deep learner. Moreover, to exploit different modalities in the selection of biomarkers, we will adopt a mixture-of-experts procedure for the integration.

Task 4.2. To decrease noise in omics signals, we will follow a within-sample normalisation strategy that we pioneered for NIPT analyses [5]. Here, we will use an autoencoder strategy with an attention mechanism to learn sample-specific normalisation parameters for the biomarkers of interest.

Task 4.3. Omics-based biomarkers are related to physiology as measured in the microfluidic chips using machine learning techniques to predict omics readouts from the physiological parameters. When trained these models can then be deployed to select cells only on their physiology (alleviating the need to measure omics).

Task 4.4. When machine learning aimed at predicting physiology is feasible on microfluidic chip- (WP1/2) and omics-based readouts the cost effectiveness of the technology relevant disease setting will be evaluated to determine the potential value for society early-on.

WP5. To build a private-partnership community.

Participants: Martens (lead) together with all participants.

Task 5.1. We will build a Dutch Society for Liquid Biopsy diagnostics and connect to European and global networks of scientists to better position the emerging field of liquid biopsy-based diagnostics.

Task 5.2. We will organise scientific hackathons inviting international experts directed to creating novel research and translational ideas.

Task 5.3. We will create a Convergence entrepreneurial ecosystem on liquid biopsy connecting scientists with companies and startups. To further maximize economic value created, we will also engage SMEs and large diagnostic companies.

WP6. To develop educational programs for the next generation (engineers, biologists, clinicians).

Participants: Martens (lead) together with all participants.

Task 6.1. We will extend our teaching effort and develop elective courses in the nanobiology curriculum (TUD/EMC), the Medicine curriculum (EMC) and in the Erasmus School of Health Policy and Management (EUR). We will also develop a course on liquid biopsy technology in the new postgraduate Biomolecular Sciences. Details are outlined in Ad. 2d.

WP7. To manage the Flagship, to valorize results and reach out to society.

Participants: Martens (lead) together with all participants.

Task 7.1 To organise monthly WP meetings, annual SAB meetings.

Task 7.2 write prominent scientific publications and annual scientific and financial reports

Task 7.3. Protect intellectual property and valorise output.

Details of the management and governance are outlined in Ad 4.e

SMART milestones below clarify what the project will deliver:

Work package deliverables	22	2023	2024	2025	2026	2026
D.1.1A prototype microfluidic platform to isolate, using DEP forces, rare cells from body fluids						
D.2.1A prototype microfluidic platform to isolate EVs based on antibody capture and physical properties						
D.2.2 Measure biophysical properties of health and disease-derived EVs						
D.2.3 Nanoscale fingerprinting EV proteins at single-molecule level						
D.3.1 Assays to characterise enriched cells at multi-omics levels						
D.3.4 Cost-effectiveness analysis for the implementation of each of the proposed assays						
D.4.1 Computational methods to integrate biophysical and molecular multi-layered data						
D.5.1 Match-making events between scientists, entrepreneurs, CEOs of SMEs and larger diagnostics						
D.5.2A community with expertise to process liquid biopsy research						
D.6.1 An elective courses and minors on liquid biopsy technology @ TU-Delft/ErasmusMC						
D.7.1 To write scientific papers, perspectives, reviews and report on output from the flagship						

3.d Alignment of the program with relevant national and international research agendas

This Flagship aligns with the national scientific agenda (NWA) route "Personalised medicine, health care research, sickness prevention and treatment". More specifically our aim – to develop innovative diagnostics to assess health and disease in a minimally invasive manner matches well with NWA research question 95: "How can we personalise health care, for example by using biomarkers?". The devices and associated assays to be developed match with NWA route: "Materials – Made in Holland" and NWA route: "Measuring and detecting: anything, anytime, anywhere". We align with the Horizon Europe Call (Horizon-CL4-2022-Digital-Emerging-01-19) on 2D materials-based devices and systems for biomedical applications. We will create a national hub of the European Liquid Biopsy Society to which some of the participants are already engaged.

4. Consortium (25%) max 2000 words for 4.a to 4.d

4.a Track record of the consortium members

In total, 10 investigators from 4 departments from Erasmus MC, 5 departments from TU-Delft and one from the EUR are engaged and below the track record of individual PIs are specified.

Prof. dr. Martens (EMC) (PhD, Molecular Biology, 1994) works at the forefront of translational cancer genomics research. Using liquid biopsies, he studies the evolution towards therapy resistance. He received funding from the Dutch cancer society, NWO and various EU programs. He has supervised 5 postdocs (PDs) and 25 PhDs (8 ongoing) and >25 MSc/BSc students.

Dr. Pouyan Boukany (TUD) (PhD, Polymer Science, 2008) focuses on fundamental and applied topics in living soft matter, with a major emphasis on controlling and understanding cell migration and DNA transport, the dynamics of biomolecules at micro/nanoscale. He received several prestigious grants (Marie-Curie fellowship/ERC-StG/ERC-CoG).

He has supervised 8 PDs (2 ongoing) and 9 PhDs (5 ongoing) and >50 Msc/Bsc students.

Prof. dr. Gijsje Koenderink (TU Delft, PhD Physical chemistry, 2003) is an expert in single-cell biomechanics who addresses together with biological/biomedical groups the role of mechanics in disease and tissue regeneration. She is a Young KNAW Academy member (2008) and a NWO Round Table Physics member who received prestigious grants (VIDI/VICI/ERC-StG) and the Dresden Physics Prize (2020). Since 2006, she has supervised 17 postdocs (3 ongoing), 18 PhD students (5 ongoing) and >80 undergraduate students.

Dr. Sabina Caneva (TU Delft, PhD, Engineering, 2016) is an expert in 2D materials and nanodevices applied to single-molecule biophysics. Her independent research group develops novel nanoelectromechanical systems for molecular diagnostics using a combination of nanopore measurements, fluorescence imaging and acoustofluidics. She received prestigious grants including a Marie Curie fellowship (2017) and an ERC StG (2022)

She is currently supervising 1 postdoc, 1 PhD student and 4 MSc students.

Dr. Mastrangeli (TUD) (PhD, Materials Engineering, 2010) is an expert in micro/nanofabrication and design of organ-on-chip devices with integrated microsensors and microactuators for disease modelling and pre-clinical drug testing. His papers received >2200 citations (i.e. Nature, Nature Nanotechnol, Science Adv). He is co-applicant in TTW Perspectief and NWA-ORC grants.

Since joining TUD he supervised 11 MSc (5 on going) and 24 BSc thesis projects.

Prof. dr. Gribnau (EMC) (PhD, gene expression & chromatin remodelling, 1999) is head of the department of Developmental Biology (Erasmus MC) and Oncode Investigator. His research is aimed at providing a better understanding of epigenetic mechanisms in regulation gene expression and translation of these insights into the clinic. He is an EMBO member and the recipient of prestigious grants (VIDI/VICI/ERC/Huygens-Descartes prize).

Prof. dr. Guido Jenster (EMC, PhD Molecular Endocrinology, 1994) is a leader in the field of prostate cancer, EV-research and genomics (cofounder NLSEV). He coordinates various consortia

(IMMPROVE (KWF) & proEVLifeCycle (ITN)), developed EV detection systems (EVQuant, TR-FIA (licensed)), and performed translational omics biomarker research on liquid biopsies. Since 2007, he supervised 23 PhD, 29 MSc, 23 BSc and 7 visiting scientists.

Prof. dr. Roland Kanaar (EMC, PhD, DNA recombination, 1988) is the head of the department of Molecular Genetics and an Oncode Institute Investigator. He is an elected member of the Royal Netherlands Academy Arts and Sciences (KNAW). His group investigates the molecular mechanisms and physiological role of the DNA damage response in the context of cancer.

Dr. Menno Creyghton (EMC, PhD, Cancer genetics, 2006) is an expert in epigenomics studying cell state changes during development as well as in disease. His group is technology-oriented, aiming to increase the resolution of combined large scale transcriptomic and epigenomic analyses in single cells and to enhance our understanding of disease-associated cell states. He received grants (KWF/Parkinson's & ALS foundations). He has supervised 2 Postdocs and 6 PhD students.

Prof. dr. Marcel Reinders (TUD) (PhD Electrical Engineering, 1994) is Professor in Bioinformatics and Computational Life Sciences at LUMC. He is scientific director of the AI for Biosciences lab and the Delft Bioengineering Institute, member the Data Analytics Research Facility at LUMC, and SAB member of the Dutch Techcentre for Life Sciences (DTL), scientific director of the Netherlands Bioinformatics Center (NBIC).

He has supervised >10 PDs, 56 PhD students (15 ongoing), >125 MSc students, and has 12 U(H)Ds in his group(s).

Prof.dr. Maarten IJzerman (EUR) (PhD, Biomedical sciences, 1998) is incoming Dean for the Erasmus School of Health Policy and Management and a global expert in the implementation of genomics enabled precision medicine, simulation modelling, and clinical decision support. He leads international research programs on the clinical and economic value of complex genomic sequencing including analysis of liquid biopsies. He is the

Additionally, a wide variety of clinical and translational disciplines are affiliated to this Flagship (their clinical or technical expertise is listed below in 4c) enabling us to test developed technologies in relevant disease settings.

4.b Expertise of the consortium members in relation to the aims and objectives of the program

The program lead (**Martens**) has twelve years of experience in the development and application of blood-derived liquid biopsies in translational research in all main cancer types and has extensive expertise in various types of genomics in the field of solid tumours (mainly breast cancer). **Martens** is embedded in a clinical department and has access to clinical specimens needed for the tools being developed in the Flagship via an on-going and approved liquid biopsy biobanking protocol. **Jenster** is similarly engaged in liquid biopsy research for many years but focuses on urothelial cancer, with urine as a substrate and exosomes as an analyte. **Boukany** brings expertise in the isolation of various circulating and rare cell types based on their distinct physicochemical characteristics from blood

cells, and single cell biophysical characterization and imaging. TUD will provide the cleanroom technologies for the fabrication of microfluidics with precise spatiotemporal control to isolate rare cells and handle 3D cell culture conditions. Boukany lab is embedded in the Chemical Engineering Department aimed at translating molecular understanding into health-care products and technologies. **Mastrangeli** provides expertise on micro/nanofabrication, design and fabrication of organ-on-chip devices with integrated microsensors and microactuators. **Koenderink** provides expertise on single-cell mechanical phenotyping correlated with microscopy-based quantitative analysis of cytoskeletal and nuclear organisation. **Kanaar's** group will devise functional assays for candidate markers identified from the liquid biopsies that may guide potential therapeutic options. **Caneva** is interested in bridging advances in instrumentation and nanofabrication with insights into biophysical phenomena at the smallest scale. Her lab will contribute novel ultrasensitive and high throughput single-molecule platforms in WP2 to detect and characterise disease-derived extracellular vesicles and macromolecules directly derived from plasma. **Creyghton/Gribnau** provide expertise on single cell multi omic analysis using combinatorial indexing and robotics. This will enable the development of a range of assays as specified in WP3. **Reinders** provides expertise on analysing single cell genomics data as well analysis tools of aberrations in cell-free DNA. His algorithm, WISECONDOR (Nucleic acids research, 2014), is being used by hospitals throughout the world to detect chromosomal aberrations (e.g. trisomy pregnancies) in foetal DNA present in maternal plasma (NIPT test; TRL9). **Gribnau** and **Martens** have developed MeD-Seq for liquid biopsies, which may prove a universal marker for monitoring disease in cancer and possibly beyond. Finally, **IJzerman** provides his expertise on global implementation of genomics enabled precision medicine.

Overall this team is highly suited to kick-start this innovative Flagship, which aims to revolutionise disease monitoring using rare analytes in liquid biopsies.

4.c Involvement of partners

This consortium focuses on integrated analysis pipelines and new technologies to characterise body fluids using fundamental insights in the molecular and cellular biology of disease as a guiding principle. Many researchers within Erasmus MC and TU Delft with technical and/or molecular expertise have already shown interest in our Flagship and have expressed an interest to become involved as the Flagship develops further:

- dr. Dimpna Meijer, Bionanoscience (TUD): Cryo-EM imaging of EVs in Neurology & oncology
- dr. Volkert van Steijn, Process Engineering (TUD): Microfluidics for EV separation
- dr. Farbod Alijani, 3ME (TUD): Micro- and nanosystems
- dr. Martin van Royen (Pathology (EMC): EV quantitation and characterization
- dr. Kateřina Staňková, Technology, Policy & Management (TUD): Dynamic Game Theory
- dr. Rienk Eelkema, Chemical Engineering (TUD): Amino acid labelling methods
- dr. Elena Martens-Uzunova, Urology (EMC): Epi-transcriptomics and small RNAs
- dr. Wilfred van IJcken (EMC): genomics core facility
- dr. Jeroen Demmers (EMC): proteomics core facility

In addition, to test and implement our developed technology, we are connected through a variety of affiliates from the clinical arena including:

- dr. Bianca Mostert, Agnes Jager and Martijn Lolkema; Medical oncology,
- prof, Jan IJzermans, Surgery,
- prof Chris Bangma, Urology,
- dr. Jose Hardillo, ENT,
- dr. Jolanda Wentzel, Cardiology,
- dr. Jorie Versmissen, dr. Monique Mulder, Internal medicine,
- Prof. Mickel Peppelenbosch, Gastroenterology,
- Prof. Luc van der Laan, Hepatology,
- Prof. John Hays, Medical Microbiology and Infectious Diseases,
- dr. Hieab Adams and dr. Pier Mastroberardino Neurodegenerative diseases () and
- Dr. Kavoussi, Epidemiology/ERGO.
- Prof. Sieger Leenstra & dr. Martine Lamfers, Neurosurgery
- Dr. Pim French, Neuro-oncology

The Flagship is affiliated to national, European and International initiatives and societies working on Liquid biopsies: Martens is an active member of the European Liquid Biopsy Society (headed by Klaus Pantel) and taking part in the Dutch initiative COIN (headed by Gerrit Meijer), which aims to implementation of ctDNA analyses clinically in the Netherlands. Jenster is co-founder and actively involved in the Netherlands Society for Extracellular Vesicles (headed by Marca Wauben).

Finally, the Flagship is connected to diagnostic companies working in the field of Liquid biopsies including Menarini (co-funding our PICTURES project), VyCAP, Roche diagnostics, Illumina, Thermo Fisher, Lighthouse Dx, ECsens, Exovectory, Besi, Micronit, Lionix, Lumicks and ExoDx.Besi, Philips, Nexperia, Methyloomics, Twist BioScience. Exovectory, Lumicks, GenomeScan and Cyclomics expressed their immediate interest in our flagship by providing a letter stating as such.

4.d Governance, organization and collaboration strategy of the consortium

The Flagship will be managed by the main leader of the consortium who has been PI or partner in several large national or international consortia. Consortium-wide decisions are made by the Executive Board (EB), which consists of the leader of the Flagship and the other core applicants of the Flagship. The EB will have a quarterly meeting or conference call to discuss progress and coherency of the project. There will be biannual meetings with all consortium members and also monthly meetings for each WP separately. We will install two advisory committees: i) a Scientific Advisory Board (SAB) to globally direct our consortium most optimally towards our goals, and ii) a User Committee (UC) to provide input on utilisation of the project's results, and dissemination to maximise the scientific and societal impact.

The following experts have agreed to be a member of the SAB:

1. Prof. Dr. Leon Terstappen, TU Twente, NL; Expert on circulating tumor cells.
2. Prof. Dr. Francesca Demichelis, Vice-Rector for research, University of Trento
3. Prof. Dr. Lodewijk Wessels, Netherland cancer Institute. Expert on data integration

The UC consists of all clinical collaborators (listed in 4.c), representatives of societal partners (e.g. patient advocates) and industrial parties interested in the outcome of the project. When diagnostics developed in this Flagships move toward clinical utility, we will involve regulatory bodies (EMA) and a member of the advisory board in molecular diagnostics (cieBOD) to discuss implementation.

At least once a year, a one-day consortium meeting will be organised by the EB for all involved senior and junior researchers, the UC, and the SAB. During this meeting, scientific results will be presented and strategies will be discussed. Dissemination and utilisation opportunities will be explicitly put on the agenda of this meeting.

Day-to-day execution of the work packages is managed by the WP-leaders in collaboration with the PIs that are involved in the respective WPs. There will be quarterly meetings of the contributors to each of the work packages. For defined sub-projects spanning multiple WPs, extra meetings will be organised when necessary.

The project leader will be responsible for the organization of consortium and EB meetings, the financial management, reporting, and other managerial duties as well as the sustainability of the Flagship.

Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table)

A pluriannual plan will be required if your proposal is accepted

Budget table attached: Underneath funding of projects related to proposal in the first 2 years specified

Title project	Funding body	Specification	Budget
TMCCPT: marker independent isolation cancer cells	NWO-Zon MW	1 Phd 2 jaar	€ 115,248.-
BRAINSCAPES, , single cell analyses	NWO Gravity	1 Phd 2 jaar	€ 115,248.-
PAVE, Multi-omics in CSF of Pancreatic Cancer"	Marie-Curie grant	1 Phd 2 jaar	€ 115,248.-
ABOARD "Personalized Medicine for Alzheimer's"	NWO-Zon MW	1 Phd 2 jaar	€ 115,248.-
TIMID: Taxonomy of inflammatory diseases" (proteomics and RNAseq analyses in blood	Health-Holland	1 Phd 2 jaar	€ 115,248.-
Liquid biopsy analysis in the SONIA breast cancer trial	KWF	1 Phd 1 jaar	€ 57,624.-
PICTURES: Single CTC genomics in prostate cancer	NWO-KWF LSH	1 Phd 1 jaar	€ 57,624.-
TMCCPT marker independent isolation cancer cells	Zon MW	Technician	€ 60,946.-
Liquid biopsy analysis in the SONIA breast cancer trial	KWF	Materials	€ 50,000.-
PALLETTE: CTC genomics in breast cancer	Pfizer	Materials	€ 50,000.-
Matching provided in total (i.e. in cash)			€ 851,488.-
Matching requested in total (i.e. in kind)			€ 754,792.-

II Letters of intent and letters of commitment

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.

Four Letters attached

III Letter of support from department head of the main lead

On headed paper of the department or institution.

Letter attached

IV References

1. Alix-Panabières C, Pantel K. Liquid Biopsy: From Discovery to Clinical Application. *Cancer Discov.* 2021 Apr;11(4):858-873.
2. Beijer N, Martens JWM, Sleijfer S. Incorporating liquid biopsies into treatment decision-making: obstacles and possibilities. *Drug Discov Today.* 2019 Sep;24(9):1715-1719
3. Salman JW, Schoots IG, Carlsson SV, Jenster G, Roobol MJ. Prostate Specific Antigen as a Tumor Marker in Prostate Cancer: Biochemical and Clinical Aspects. *Adv Exp Med Biol.* 2015;867:93-114.
4. Köhn L, Johansson M, Grankvist K, Nilsson J. Liquid biopsies in lung cancer-time to implement research technologies in routine care? *Ann Transl Med.* 2017 Jul;5(13):278.
5. Mokveld T, Al-Ars Z, Sistermans EA, Reinders M. WisecondorFF: Improved Fetal Aneuploidy Detection from Shallow WGS through Fragment Length Analysis. *Diagnostics (Basel).* 2021 Dec 28;12(1):59
6. Liu Q, Muralidharan A, Saateh A, Ding Z, Ten Dijke P, Boukany PE. A Programmable Multifunctional 3D Cancer Cell Invasion Micro Platform. *Small.* 2022 Mar 9:e2107757.
7. Meijer TG, Verkaik NS, Sieuwerts AM, van Riet J, Naipal KAT, van Deurzen CHM, den Bakker MA, Sleddens HFBM, Dubbink HJ, den Toom TD, Dinjens WNM, Lips E, Nederlof PM, Smid M, van de Werken HJG, Kanaar R, Martens JWM, Jager A, van Gent DC. Functional Ex Vivo Assay Reveals Homologous Recombination Deficiency in Breast Cancer Beyond BRCA Gene Defects. *Clin Cancer Res.* 2018 Dec 15;24(24):6277-6287.

8. Martens JW, Sieuwerts AM, Bolt-deVries J, Bosma PT, Swiggers SJ, Klijn JG, Foekens JA. Aging of stromal-derived human breast fibroblasts might contribute to breast cancer progression. *Thromb Haemost.* 2003 Feb;89(2):393-404
9. S.H. Hilton, I.M. White. Advances in the analysis of single extracellular vesicles: A critical review. *Sensors and Actuators Reports*, Volume 3, November 2021, 100052
10. T.A. Hartjes, J.A. Slotman, M.S. Vredenburg, N. Dits, R. Van der Meel, D. Duijvesz, J.A. Kulkarni, P.J. French, W.A. Van Cappellen, R.M. Schiffelers, A.B. Houtsmuller, G.W. Jenster, M.E. Van Royen. EVQuant; high-throughput quantification and characterization of extracellular vesicle (sub)populations. *bioRxiv*: doi.org/10.1101/2020.10.21.348375
11. Maria A Lebedeva, Elena Palmieri, Philipp Kukura, Stephen P Fletcher. Emergence and Rearrangement of Dynamic Supramolecular Aggregates Visualized by Interferometric Scattering Microscopy. *ACS Nano.* 2020 Sep 22;14(9):11160-11168. doi: 10.1021/acsnano.0c02414
12. Alfaro A. J. et al., The emerging landscape of emerging single-molecule sequencing technologies. *Nature Methods*, 18, 604-617 (2021)
13. B. Mateescu *et al.*, Obstacles and opportunities in the functional analysis of extracellular vesicle RNA – an ISEV position paper. *JEV*, 2017, Vol. 6, 1286095
14. Chatalic KL, Veldhoven-Zweistra J, Bolkestein M, Hoebe S, Koning GA, Boerman OC, de Jong M, van Weerden WM. A Novel ¹¹¹In-Labeled Anti-Prostate-Specific Membrane Antigen Nanobody for Targeted SPECT/CT Imaging of Prostate Cancer. *J Nucl Med.* 2015 Jul;56(7):1094-9.
15. Long HK, Prescott SL, Wysocka J. Ever-Changing Landscapes: Transcriptional Enhancers in Development and Evolution. *Cell.* 2016 Nov 17;167(5):1170-1187
16. Lee KCM, Guck J, Goda K, Tsia KK. Toward Deep Biophysical Cytometry: Prospects and Challenges. *Trends Biotechnol.* 2021 Dec;39(12):1249-1262